

The circulatory system

- Transports fluids throughout the body.
- consists of two interrelated parts,
- which function in parallel to transport the body's fluids:

1. Cardiovascular system

2. Lymphatic system

Cont...

- The heart and blood vessels make up the **blood** transportation network,
 - the cardiovascular system
- The lymphatic system, consists of
 - lymph vessels
 - lymph nodes
 - Lymphatic organs and lymph

Functions of the circulatory system

- Distributes nutrients
- Transport and exchange of oxygen and carbon dioxide
- Removal of waste materials
- Distributes secretions of endocrine glands
- Prevent excessive bleeding
- Prevent infection
- Regulate body temperature.



Major Parts of the Cardiovascular system

- **Blood Vessels** - routes of blood travels
- **Heart** - pumps or pushes blood through body
- **Blood** - is liquid connective tissue

Function of blood

- Transportation
- Regulation
- protection

Blood

- 5-6 liters in average adult male and 4-5 liters in adult female.
- The difference in volume is due to differences in body size.
- Constituted about 8% of the total body weight
- Its pH range from 7.35 to 7.45.

Components of Blood

- Whole blood is composed of two portions:

1. Blood plasma

- liquid extracellular matrix that contains dissolved substances.

2. Formed elements

- which are cells and cell fragments.
- Blood is about 45% formed elements and 55% plasma.
- more than 99% of the formed elements are red blood cells (RBCs).

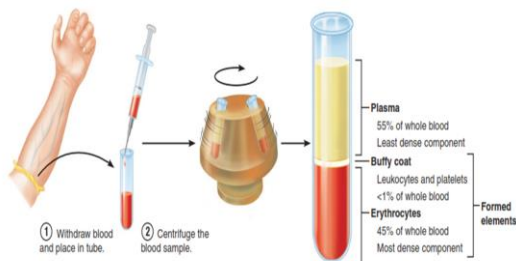
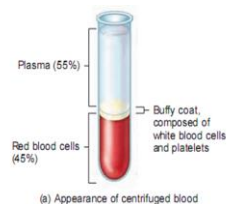


FIGURE 18.1 Major components of whole blood.

Cont...

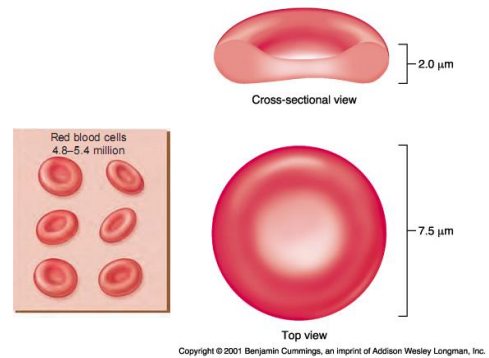
- The percentage of total blood volume occupied by RBCs is termed the **hematocrit**.
- WBCs and platelets occupy less than 1% of total blood volume.
- form a very thin layer, called the **buffy coat**.



Red blood cells

- Biconcave discs-discs with depressed centers
- mature RBC **lacks** nucleus and other **organelles**
- their cytoplasm is packed with molecules of **hemoglobin**- the oxygen carrying protein.
- erythrocytes are over 97 % hemoglobin.
- live (life span) **100-120** days much longer than other types of blood cells.
- **Erythropoiesis** is the production of RBCs starts in the red bone marrow.
- function-transport O_2 & CO_2

Cont...



White blood cells (Leukocytes (WBC))

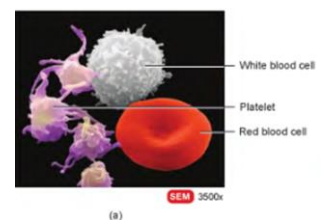
- Unlike red blood cells, white blood cells have nuclei and do **not** contain **hemoglobin**,
- that is why they are colorless.
- less numerous than erythrocytes
- can move in **ameboid** fashion and protects the body from **infectious** microorganisms such as bacteria, virus and parasites
- can function outside the blood stream
- originate from bone marrow

WBCs are classified as:

1. Agranulocytes
2. Granulocytes

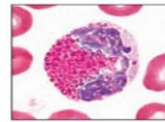
- Depending on whether they contain or not **conspicuous chemical-filled** cytoplasmic **granules** that are made visible by staining when viewed through a **light microscope**.

Cont...

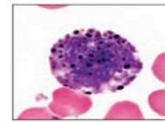


Granulocytes (granular leukocytes)

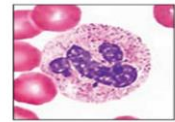
- There are 3 types of granulocytes.
- Neutrophils, eosinophils , basophiles
- are larger and much shorter lived than erythrocytes.
- Functionally , all granulocytes are phagocytic ;
 - they engulf and digest foreign cells or molecules.
- Have variable shaped nuclei.



Eosinophil



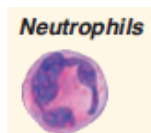
Basophil



Neutrophil

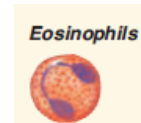
Neutrophils

- 2-5 lobed nucleus
- the most abundant type of leukocytes (constitutes 60%)
- Neutrophils respond first to bacterial invasion.
- Destroy bacterial



Eosinophils

- accounts for 1%-4% of all leukocytes
- bilobed nucleus
- phagocytize antigen-antibody complexes
- are effective against certain parasitic worms.



Basophiles

- accounts 0.5–1% of all WBCs.
- bilobed nucleus
- release histamine and other mediators of inflammation.



Agranular leukocytes (agranulocytes)

- Have **no** granules after staining
- two types
 - lymphocytes
 - monocytes

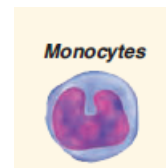
Lymphocytes

- 20-45% of all leukocytes
- are effective in fighting **infectious** organisms
- types: B cells and T cells



Monocytes

- 12–20 micro m. diameter;
- The **largest** leukocytes.
- makes up 4-8% of WBC
- phagocytosis after transforming into fixed or **wandering** macrophages.



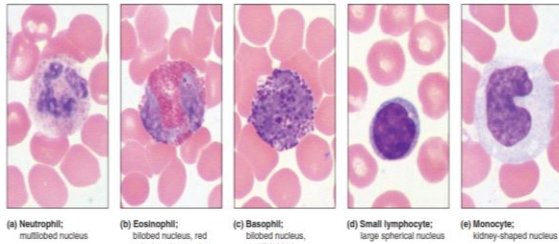


FIGURE 18.4 Leukocytes. In each case, the leukocytes are surrounded by erythrocytes. (All 1700 $\times$, Wright's stain.) (See *A Brief Atlas of the Human Body*, Second Edition, Plates 23–27)

TABLE 18.1 Summary of Formed Elements of the Blood

Cell Type	Illustration	Description*	Number of Cell per mm ³ (μ L) of Blood	Duration of Development (D) and Life Span (LS)	Function
ERYTHROCYTES (red blood cells; RBCs)		Biconcave, anucleate disc; salmon-colored; diameter 7–8 μ m	4–6 million	D: 5–9 days LS: 100–120 days	Transport oxygen and carbon dioxide
LEUKOCYTES (white blood cells, WBCs)		Spherical, nucleated cells	4800–11,000		
Granulocytes					
• Neutrophils		Nucleus multilobed; inconspicuous cytoplasmic granules; diameter 12–14 μ m	3000–7000	D: 7–11 days LS: 6 hours to a few days	Destroy bacteria by phagocytosis
• Eosinophils		Nucleus bilobed; red cytoplasmic granules; diameter 12–15 μ m	100–400	D: 7–11 days LS: about 5 days	Turn off allergic responses and kill parasites
• Basophils		Nucleus bilobed; large blue-purple cytoplasmic granules; diameter 10–12 μ m	20–50	D: 3–7 days LS: a few hours to a few days	Release histamine and other mediators of inflammation

Table 17.1 Summary of Formed Elements of the Blood

Cell Type	Illustration	Description*	Number of Cells/mm ³ (μ L) of Blood	Duration of Development (D) and Life Span (LS)	Function
• Basophils		Nucleus bilobed; large blue-purple cytoplasmic granules; diameter 10–12 μ m	20–50	D: 3–7 days LS: 7 (a few hours to a few days)	Release histamine and other mediators of inflammation
Agranulocytes					
• Lymphocytes		Nucleus spherical or indented; pale blue cytoplasm; diameter 5–17 μ m	1500–3000	D: days to weeks LS: hours to years	Mount immune response by direct cell attack (T cells) or via antibodies (B cells)
• Monocytes		Nucleus U- or kidney-shaped; gray-blue cytoplasm; diameter 14–24 μ m	100–700	D: 2–3 days LS: months	Phagocytosis; develop into macrophages in tissues
Platelets		Discoid cytoplasmic fragments containing granules; stain deep purple; diameter 2–4 μ m	250,000–500,000	D: 4–5 days LS: 5–10 days	Seal small tears in blood vessels; instrumental in blood clotting

*Appearance when stained with Wright's stain.
Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.

Platelets (Thrombocytes)

- the **smallest** formed elements.
- 2–4 micro.m diameter cell fragments.
- are fragments of large cells called megakaryocytes.
- have **no** nucleus.
- capable of ameboid movement
- blood **clotting**.

Platelets (Thrombocytes)



Heart

- Hollow, muscular, 4 chambered organ
- Weighs about 350 g in adults (about the size of a **clenched fist**)
- it rests on the diaphragm near the mid line of the thoracic cavity in the **mediastinum**
- propel blood to all parts of the body.
- The right side of the heart receives poorly oxygenated blood
- The left side of the heart receives well-oxygenated (arterial) blood.

❖ **The mediastinum:** is a space b/n **two** lungs

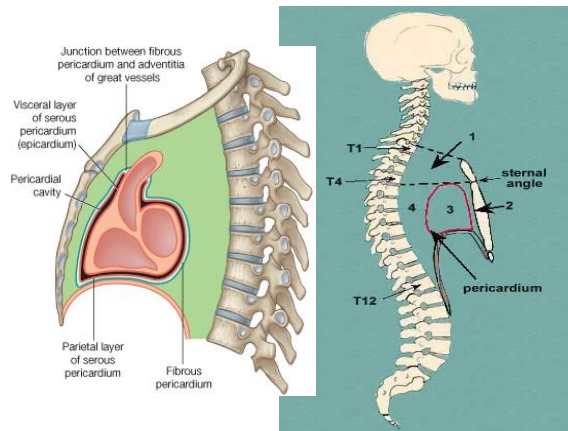
- It extends from the superior **thoracic aperture** to the **diaphragm** inferiorly and from the sternum and costal cartilages **anteriorly** to the bodies of the **thoracic vertebrae posteriorly**.
- The major structures in the mediastinum are also surrounded by **blood** and **lymphatic** vessels, lymph nodes, nerves, and fat
- It contains **all** the thoracic viscera and structures, **except** the lungs.
- The mediastinum is artificially divided into **superior** and **inferior parts** for purposes of **description**.

➤ **The superior mediastinum**

- extends inferiorly from the superior **thoracic aperture** to the **horizontal plane** (transverse thoracic plane) passing through the **sternal angle** and the **IV disc** of the **T4&T5 vertebra**
- ❖ It contains:
 - the superior vena cava (SVC)
 - brachiocephalic veins
 - arch of the aorta
 - thoracic duct
 - trachea, esophagus, thymus, vagus nerves
 - left recurrent laryngeal nerve, and phrenic N.

➤ **The inferior mediastinum**

- between the **transverse thoracic plane** and the **diaphragm**,
- It is **further subdivided** by the pericardium into :
- ✓ **The anterior mediastinum**
 - containing remnants of the thymus, lymph nodes, fat, and connective tissue
- ✓ **The middle mediastinum**
 - containing the pericardium, **heart**, roots of the great vessels, arch of **azygos** vein, and main **bronchi**
- ✓ **The posterior mediastinum**
 - containing the **esophagus**, **thoracic** aorta, azygos and **hemiazygos** veins, thoracic duct, **vagus** nerves, **sympathetic** trunks, and **splanchnic** nerves



Structure of the heart

❖ Pericardium (peri = around)

- is a protective and surrounding membrane of the heart
- it consists of two principal portion:
 - the **fibrous** pericardium
 - the **serous** pericardium

➤ Fibrous pericardium

- is tough inelastic, dense irregular connective tissue, which looks like a **bag**.
- It rests & attaches to the diaphragm:

- Functions of fibrous pericardium

- prevents **overstretching** of the heart
- provides **protection**
- **anchors** the heart in the mediastinum

➤ The serous pericardium

- is a thinner membrane which forms a **double** layer

▪ **parietal layer**

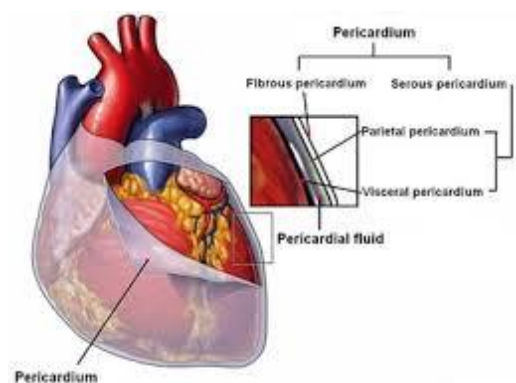
- ✓ is the outer serous pericardium

▪ **visceral layer**

- The inner serous pericardium
- adheres tightly to the surface of the heart.

❖ **pericardial fluid**

- is a fluid in the cavity b/n two layers of pericardium
- is important to reduce **friction** b/n the heart and the outer layers and also prevent the heart from **external pressure**.



❖ Layers of the heart wall

➤ The heart contains 3 distinct layers:

✓ Epicardium

- **thin**, external membrane around the heart.
- allow **protection** against friction by rubbing organs

✓ Myocardium

- **thickest** layer; consists of **cardiac** muscle. i.e. the bulk of the heart wall.
- It is **involuntary** muscle
- responsible for the ability of the heart to **contract**.

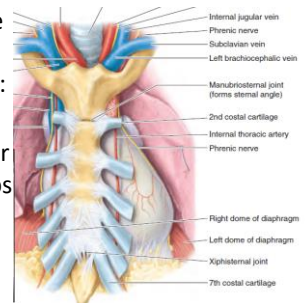
✓ Endocardium

- a thin **innermost** layer; a unique type of **epithelial** tissue that **lines** the **entire** circulatory system.

Relations and external features of the heart

- The heart within the pericardial sac is related anteriorly to:

- the sternum, costal cartilages, and anterior ends of the 3rd-5th ribs on the left side.

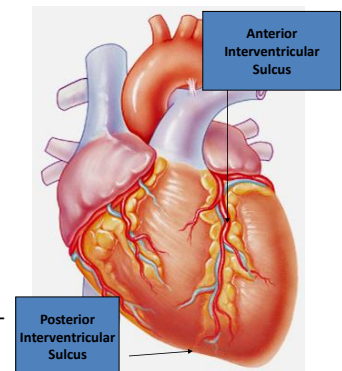


Cont...

- Externally, the atria are demarcated from the ventricles by,
 - the **coronary** or **atrioventricular** groove (sulcus).
- the right and left ventricles are demarcated from each other by,
 - anterior and posterior **interventricular** (IV) **grooves**.
- The heart has an **apex**, a **base**, four **surfaces** and four **borders**

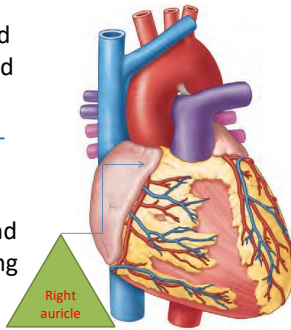
Heart Chambers

- ✓ The anterior inter-ventricular sulcus, separates the right and left ventricles
- ✓ It continues as the posterior inter-ventricular sulcus which provides a similar landmark on the heart's postero-inferior surface



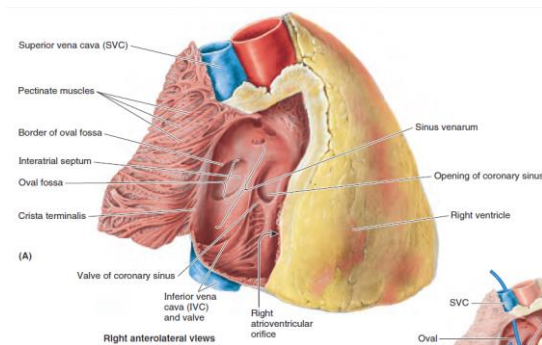
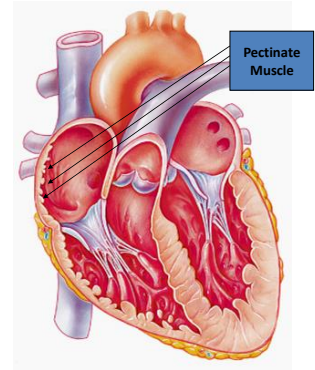
Right atrium

- forms the right border of the heart and receives **venous** blood from the SVC, IVC, and coronary sinus
- **Right auricle**- the **ear-like** conical muscular pouch that projects from this chamber and overlaps the ascending aorta.



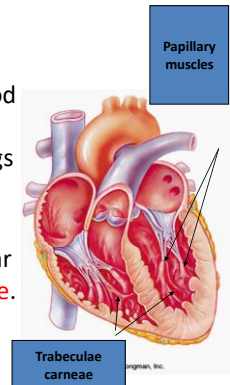
Cont...

- Internally,
 - the posterior walls are smooth,
 - but the anterior walls are ridged by bundles of muscle tissue
- ✓ These muscle bundles are called **pectinate muscles**

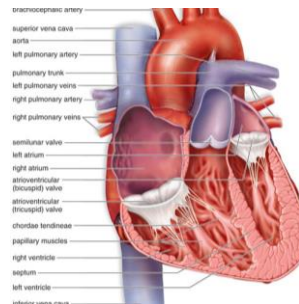


Right ventricle

- forms the largest part of the anterior surface of the heart
- The right ventricle pumps blood into the **pulmonary trunk**, which routes blood to the lungs for gas exchange.
- The interior of the right ventricle has irregular muscular elevations - **trabeculae carneae**.
- **conus arteriosus**, smooth walled outflow part.



- Cone shaped **papillary muscles** project from the walls in to ventricular cavity
- **Tendinous cords (L. chordae tendineae)** arise from the apices of papillary muscles
- and attach to the free edges and ventricular surfaces of the **anterior**, **posterior**, and **septal** cusps.

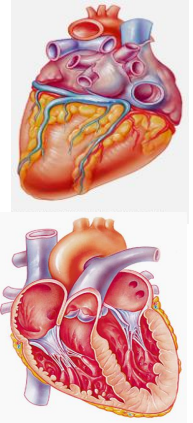


Left atrium

- Rectangular in shape
- forms most of the base of the heart.
- Smaller than the right atrium
- Blood enters the left atrium via four veins

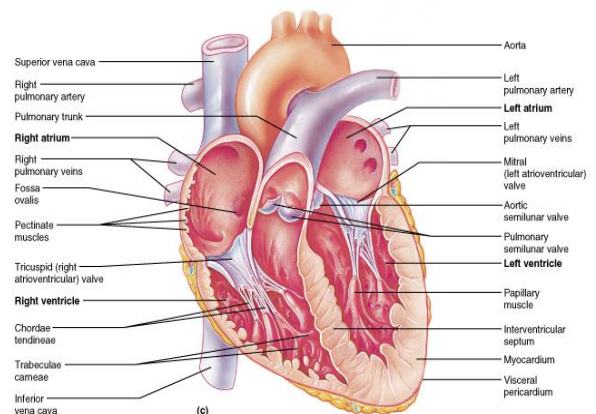
➤ **Right and left pulmonary veins**

- **Left auricle** - forms the superior part of the left border of the heart and overlaps the pulmonary trunk
- Most of the atrial wall – **smooth**



Left ventricle

- Forms the apex of the heart, nearly all its left (pulmonary) surface and border, and most of the diaphragmatic surface
- Walls that are 2-3 times as thick as that of the right ventricle.
- **Trabeculae carneae** - Finer and more numerous than the right ventricle.
- Papillary muscles that are larger than those in the right ventricle
- **The aortic vestibule** – a smooth-walled outflow part located superoanteriorly.
- A double-leaflet mitral valve that guards the left AV orifice.



Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.

Heart Valves

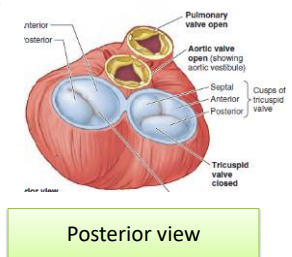
- Prevent **back** flow of blood and ensure that blood flows in the proper **direction** through the heart.
- The valves function is to maintain blood flow in **one direction**.
- Valves open and close in response to changes in **pressure**.

➤ Four heart valves are:

- Atrioventricular (AV) valves
 - ✓ tricuspid - the right AV valves
 - ✓ bicuspid valves - the left AV valves
- Semilunar valves
 - ✓ Aortic valve - Left Ventricle and Aorta
 - ✓ Pulmonary valve- Rt Ventricle and Pulmonary Trunk

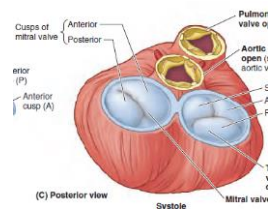
tricuspid valve

- guards the right AV orifice
- has **anterior**, **posterior**, and **septal** cusps
- Prevents backward flow of blood from the right ventricle back into the right atrium
- located posterior to the body of the sternum at the level of the **4th** and **5th intercostal** spaces



mitral valve

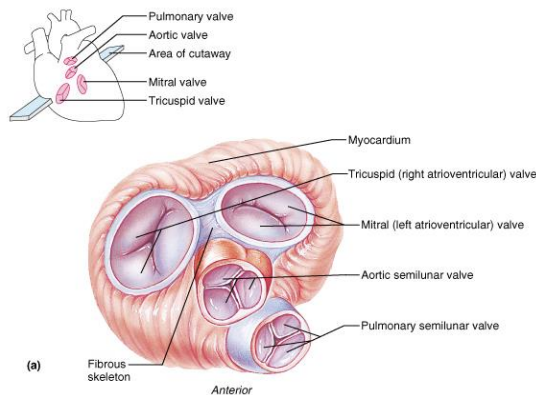
- guards the left AV orifice
- has two cusps, **anterior** and **posterior**.
- Prevents backward flow of blood from the left ventricle back into the left atrium
- located posterior to the sternum at the level of the **4th costal** cartilage.



Semilunar Valves

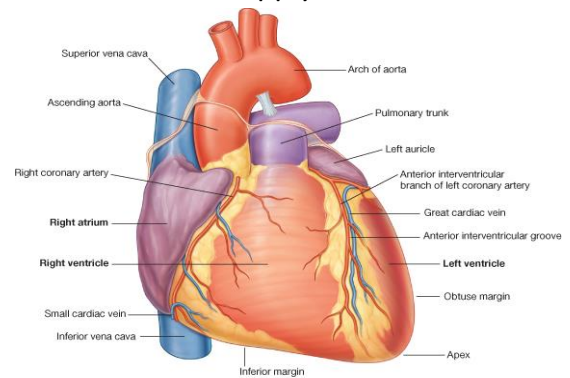
Pulmonary and aortic valves

- Each have three semilunar cusps: pulmonary valve (**anterior**, **right**, and **left**) and aortic (**posterior**, **right**, and **left**).
- Semilunar cusps do not have **tendinous cords** to support them and are smaller than the cusps of the AV valves.
- Immediately superior to each semilunar cusp, the walls of the origins of the pulmonary trunk and aorta are slightly dilated, forming a **sinus**.
- the right and the left coronary arteries arise from the right and left **aortic sinuses**, but no artery arises from the posterior aortic sinus.



Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.

Blood supply of Heart



© Elsevier. Drake et al: Gray's Anatomy for Students - www.studentconsult.com

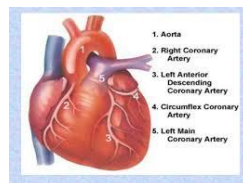
Cont'd

➤ The **heart** is supplied by two major **coronary** arteries, the **right & left coronary aa.**

✓ **Lt coronary a** divides into:

1. anterior interventricular &

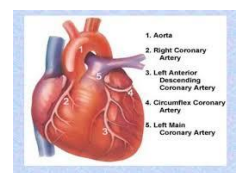
2. circumflex branches immediately after it arises from **left** side of the **ascending aorta**.



➤ The anterior interventricular a. (AIVA) lies in the **anterior interventricular sulcus** & is also known as the **Lt anterior descending a.**

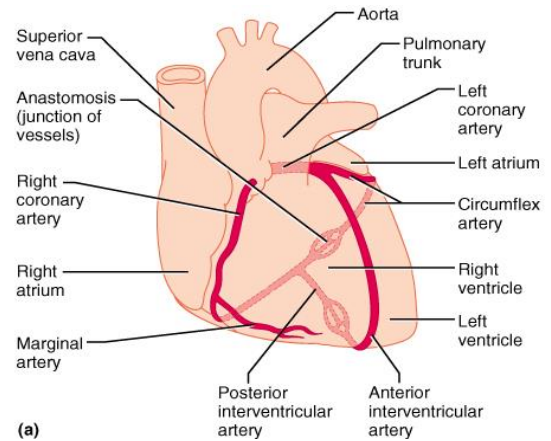
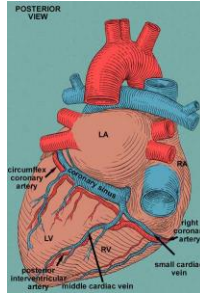
- supplies **wall of right & left ventricles, interventricular septum & apex** of the heart.

Cont'd



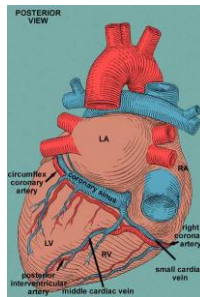
Cont'd

- The **circumflex branch** (CCA) lies in the **coronary sulcus**: Supplies walls of **LA & LV** & forms an **anastomosis** with the **right coronary a.**



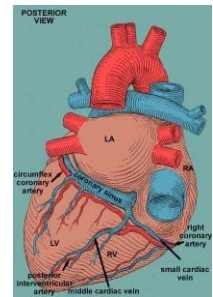
Cont'd

- ✓ **Right coronary a.** lies in the **coronary sulcus**
Branches :
- SA nodal
 - right marginal
 - posterior interventricular artery



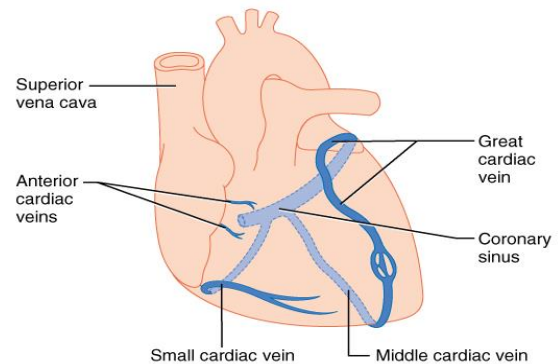
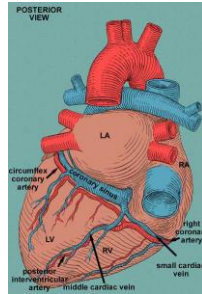
- **Right Coronary Artery** supplies **Rt atrium & Rt ventricle, sinu-atrial & atrioventricular nodes, interatrial- septum, a portion of Lt atrium, posteroinferior one-third of interventricular septum, & a portion of posterior part of Lt ventricle.**

Cont'd



- When the heart is *viewed from the back*, the most obvious structure lying in the **coronary sulcus** is the **coronary sinus**.
- **Coronary sinus** receives most of the venous blood from the heart & empties into **right atrium**.
- Its tributaries are the **small cardiac vein**, **middle cardiac vein** & **greater cardiac vein**.
- The arteries seen at the back of the heart are the **circumflex** coronary artery, terminal part of right coronary artery & its posterior interventricular branch.

Cont'd



(b)
Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.

Blood vessels

- Form a closed delivery system powered by the pumping heart.
- Are dynamic structures, that pulsate, constrict and relax according to the changing needs of the body.
- There are three types of blood vessels:
 1. Arteries
 2. Veins
 3. Capillaries.

Cont...

- Blood under high pressure leaves the heart and is distributed to the body by a branching system of thick-walled **arteries**.
- The final distributing vessels, **arterioles**, deliver oxygenated blood to **capillaries**.
- Capillaries form a capillary bed, where the interchange of oxygen, nutrients, waste products, and other substances with the extracellular fluid occurs.
- Blood from the capillary bed passes into thin-walled **venules**, which resemble wide capillaries.
- Venules drain into small veins that open into larger veins.

Cont...

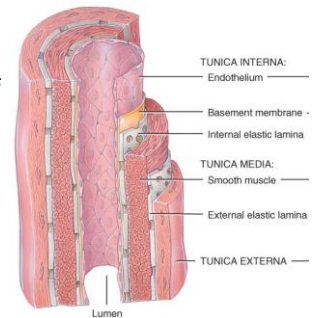
- The largest veins, the **superior** and **inferior venae cavae**, return poorly oxygenated blood to the heart.
- Arteries **branch** or diverge as they carry blood away from the heart.
- Veins **converge** or serve as tributaries as they carry blood toward the heart

Cont...

- Tunics/histological layers of blood vessels
- Wall of arteries and veins are composed of three distinct layers (tunics):

1. Tunica intima

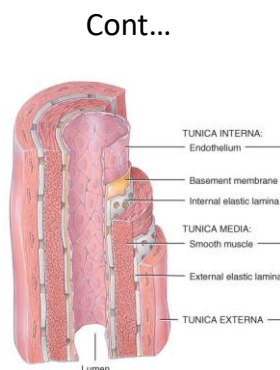
- Composed of simple, squamous epithelium called **endothelium**.



- It has **direct** contact with blood
- Rests on a CT membrane that is rich in elastic and collagen fibers.

2. Tunica media

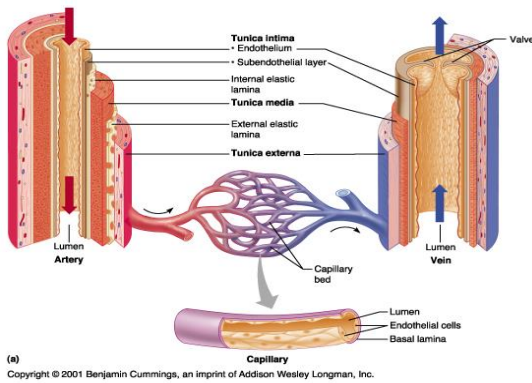
- Makes up the **bulk** of the arterial wall.
- Includes **smooth muscle fibers**, which encircle the tube, and a thick layer of **elastic** connective tissue.



Cont...

3. Tunica adventitia

- Is relatively **thin**.
- Consists chiefly of **connective** tissue with irregularly arranged **elastic** and **collagen** fibers.
- Also contains **minute** vessels that give rise to **capillaries** and provide blood to the more **external** cells of the **artery** wall.
- Capillaries have only an endothelium, with no subendothelial layer or other tunics.



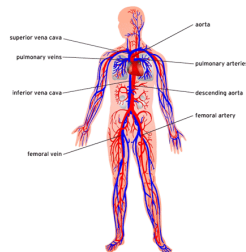
Types blood vessels

- Blood vessels are a **network** of **tubes** that carry blood throughout the entire body.
- ❖ There are **5** types of Blood Vessels:
 1. Arteries
 2. Arterioles
 3. Veins
 4. Venules
 5. Capillaries

Cont...

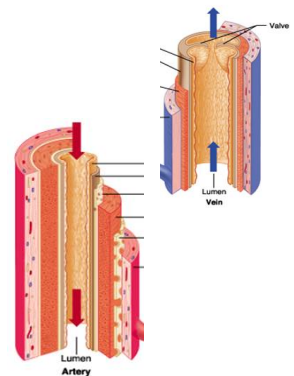
1. Arteries

- The vessels that carry the blood **away** from the heart
- All arteries carry oxygen-rich blood except the pulmonary arteries and umbilical arteries.



Cont

- Arteries have a **smaller lumen** than **veins** of similar size.
- Arterial walls are **thicker** than **venous** walls.
- Arteries have more **elastin** than veins.
- Arteries have **no** valves
- because the **blood pressure** in arteries is high enough that there is **no** backflow of blood.



Types of Arteries

A. Elastic Arteries

- Largest arteries near the heart, (2.5cm-1cm in diameter)
- Largest diameter but walls relatively **thin**
- **Function as pressure reservoir**
- Help propel blood forward while ventricles relaxing
- Also known as **conducting arteries** – conduct blood to medium-sized arteries.
- E.g. aorta, subclavian, brachiocephalic, common iliac, common carotid, pulmonary ...

Cont...

B. Muscular arteries

- Tunica media contains **more smooth** muscle and **fewer elastic** fibers than elastic arteries
- Walls relatively **thick**.
- Capable of great vasoconstriction/vasodilatation to adjust rate of blood flow.
- Also called **distributing arteries**.
- E.g. Brachial artery, radial artery, ulnar artery.

2. Arterioles

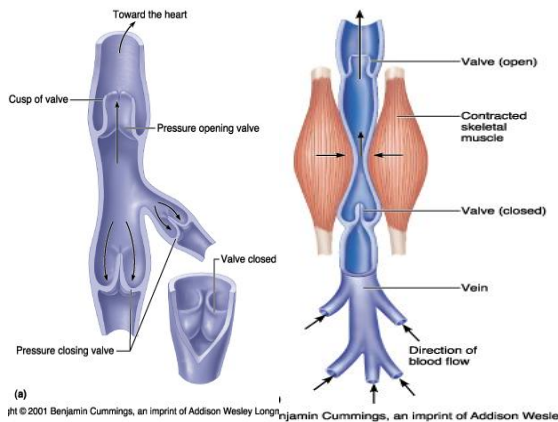
- An arteriole (small artery) is a very small, almost **microscopic**, artery that delivers blood to capillaries.
- arterioles consist of little more than a layer of endothelium
- covered by a **few** smooth muscle fibers
- **Abundant** vessels in the body

3. Veins

- Veins carry blood **towards** the heart
- carry de-oxygenated blood except the pulmonary veins.

Cont...

- Veins are more abundant than arteries.
- Although their walls are thinner, their diameters are usually larger than those of the corresponding artery.
- Because of veins' larger diameter and ability to expand, typically only 20% of the blood occupies arteries, whereas 80% is in the veins.
- In the limbs, and in some other locations where the flow of blood is opposed by the pull of gravity, veins have **valves** that permit blood to flow toward the heart but not in the reverse direction.



➤ Medium veins

- drain venous plexuses and accompany medium arteries.
- Examples of medium veins include the named superficial veins (cephalic and basilic veins of the upper limb and great and small saphenous veins of the lower limb) and the accompanying veins that are named according to the artery they accompany.

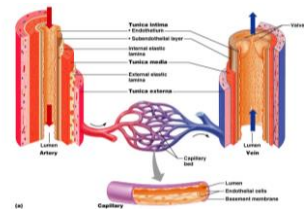
Cont...

➤ Large veins

- are characterized by wide bundles of longitudinal smooth muscle and a well-developed tunica adventitia.
- An example is the superior vena cava.

4. Venules

- Thinner walls than arterial counterparts
- Postcapillary venule smallest venule
- Form part of **microcirculatory** exchange unit with capillaries
- Muscular venules have thicker walls with 1 or 2 layers of smooth muscle



5. Capillaries

- are tiny blood vessels that pass blood from the **arteries** into the **veins**.
- They are very small, the largest being about **10** micrometers in diameter.
- Their **walls** are **thin** which allows materials to pass into the capillaries.
- the capillaries are able to **profuse** the tissues of the body with **needed** oxygen and important nutrients supplied by blood.
- **Lack** **tunica media** and **tunica externa**

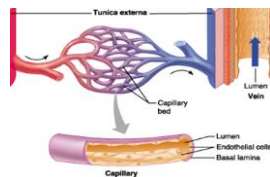
Cont...

- ✓ While capillaries function in one respect as the “communicators” b/n arteries and veins, they also are the tiny blood vessels that supply blood to organs
 - Capillaries supplying blood to an organ, when taken in whole, are called a **capillary** bed.
- In addition to being the transporters of blood products, capillaries allow for **waste** products to enter.
- In this way they perform an important function because waste is ultimately transported out of the body through this interchange.

Cont....

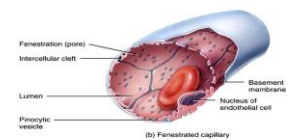
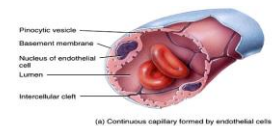
✓ Three types of capillary:

1. Continuous – most common
2. Fenestrated – have pores
3. Sinusoids (discontinuous)



1. Continuous capillaries

- have the thickest endothelial wall.
- They allow only water, and ions into their pathways.
- Mostly found in the **skeletal** & smooth muscles, **lungs**



2. Fenestrated capillaries

have “windows” that let larger molecules in and out of the capillaries.

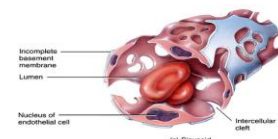


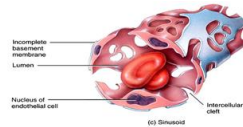
Figure 21.04 Tortora - PAP 12/e
Copyright © John Wiley and Sons, Inc. All rights reserved.

Cont...

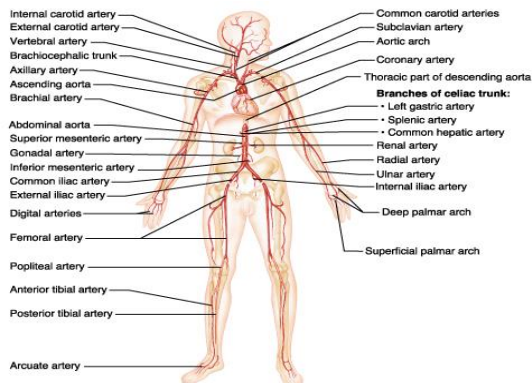
- Are found in the: **kidneys** **brain** and some endocrine glands.

3.Sinusoidal capillaries

- have the greatest amount of permeability, letting red blood cells and proteins in through the endothelial walls.



	Arteries vs	Veins
Direction of flow	Blood Away from Heart	Blood to Heart
Pressure	Higher	Lower
Walls	THICKER: Tunica media thicker than tunica externa	THINNER: Tunica externa thicker than tunica media
Lumen	Smaller	Larger
Valves	No valves	Valves



Principal Arteries of the body

❖aorta

➤ascending aorta

- ascends from the heart (left ventricle).
- The **coronary** arteries are the only branch of the ascending aorta that supplies the heart.

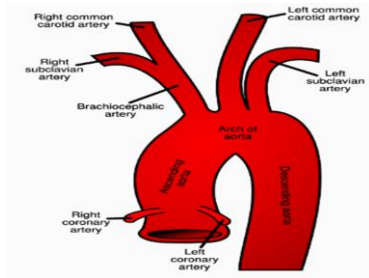
➤Aortic arch

- Three vessels arise from the aortic arch:
 - ✓ the **brachiocephalic** artery—further bifurcates into **Right** subclavian and **right** common carotid aa that supply the **right** upper limbs and head regions.

Cont...

- ✓ left common carotid artery
- ✓ left subclavian artery

➤ Descending aorta



I. Arteries of the head and neck

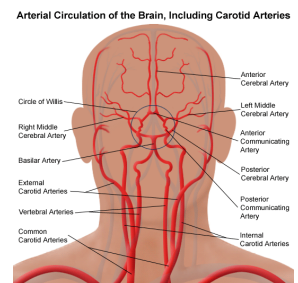
1. The brachiocephalic artery branches to form
 - Right common carotid artery-transport blood to the right side of head and neck
 - Right subclavian artery-supply blood to right upper limb
2. Left common carotid artery –branch directly from aortic arch and transport blood to the left side of head and neck.

Cont...

- ✓ The common carotid arteries at the level of mandible branch into:-
 - **external carotid artery**-have several branches that supply the structure of the **face, nose** and **mouth**
 - **Internal carotid artery**- pass through the carotid canal and supply most part of brain.
- With in the cranial vault, branches of vertebral arteries and right and left internal carotid arteries form a system of vessel called **circle of wills**.

❖ The vertebral artery

- arises from the **subclavian** artery
- ascends in the neck through the **transverse foramen**
- enters the cranial cavity through the **foramen magnum**.
- The right and left vertebral arteries unite to form the **basilar** artery.



❖ *The internal carotid arteries*

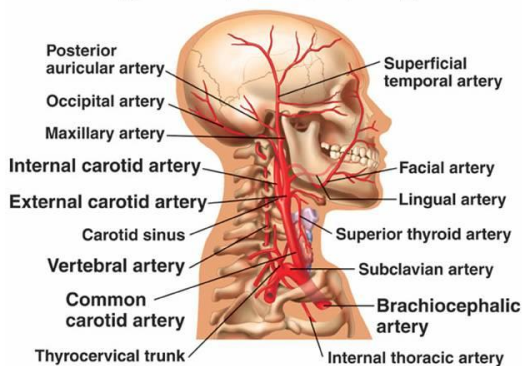
- is divided into
- the **ophthalmic** artery
 - supplies the eye, and
- the **anterior** and **middle cerebral** arteries
 - supply the cerebrum.

❖ *The external carotid*

- branches are named according to the **area** or **structures** they supply.
- **Superior thyroid artery**
 - supply hyoid, larynx, vocal cords, thyroid gland.

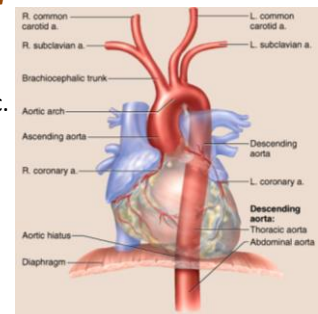
- **Ascending pharyngeal artery** - supply pharyngeal area
- **Lingual artery** - supply tongue and sublingual gland.
- **Facial artery** - supply pharyngeal, palate, chin, lips, nasal region
- **Occipital artery**
 - supply scalp (posteriorly), meninges, mastoid region, some of the posterior neck muscles.
- **Maxillary artery**
 - supply teeth, gums, muscles of mastication, nasal cavities, eyelids
- **Superficial temporal artery**
 - supply parotid gland, side of the head.

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



II. Arteries of the upper limbs

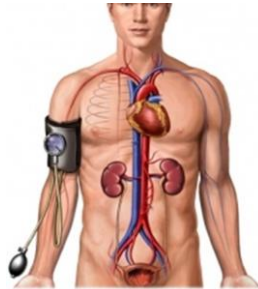
- Right subclavian
 - from brachiocephalic.
- and left subclavian
 - from aortic arch.



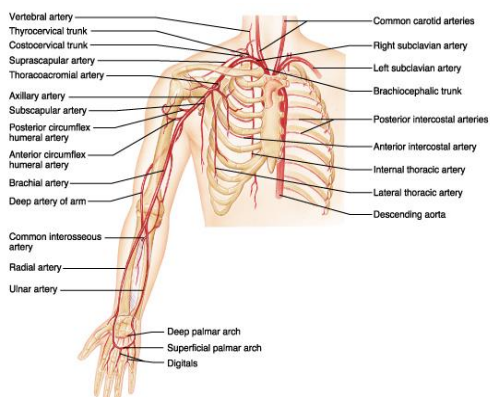
Cont...

- It has the several branches in the thorax but becomes the **axillary artery** as it passes in the **axilla**.
- It becomes the **brachial artery** in the **arm**.

➤ *Site for BP(blood pressure) measurement*

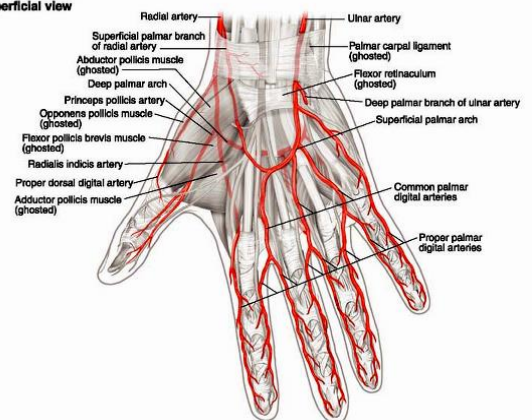


- The **brachial artery** bifurcates at the **cubital fossa** into:
 - **Radial artery**- supplies muscles on the **radial side** of the forearm. It is the site of **measuring pulse**.
 - **Ulnar artery** – supplies muscles on the **ulnar side** of the forearm.
- Both arteries form
 - **palmar arch** in the palm and digital arteries supplying the digits.



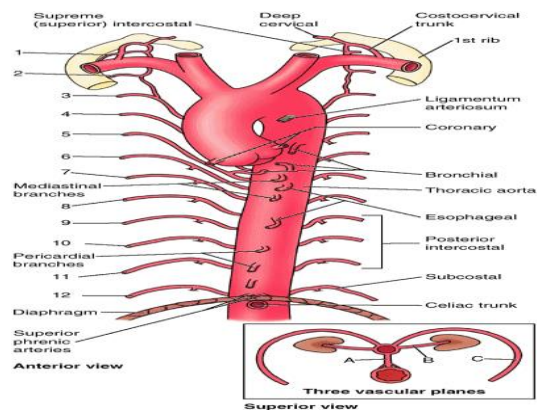
Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.

A. Superficial view



III. Thoracic Aorta

- The thoracic aorta is the continuation of the arch of the aorta
- It begins on the left side of the inferior border of the body of the T4 vertebra and terminates anterior to the inferior border of the T12 vertebra and enters the abdomen through the **aortic hiatus** in the diaphragm .
- Branches :
 - esophageal arteries
 - Bronchial arteries
 - Pericardial branches
 - Mediastineal branches
 - Posterior intercostal arteries
 - Superior phrenic arteries



IV. Abdominal Aorta

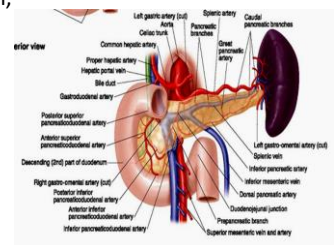
- Approximately 13 cm in length.
- begins at the aortic hiatus in the diaphragm at the level of the T12 vertebra and ends at the level of the L4 vertebra by dividing into the right and left common iliac arteries.
- each common iliac artery divides into the internal and external iliac arteries.
- The internal iliac artery enters the pelvis.
- The external iliac artery follows the iliopsoas muscle and supplies the lower limb
- Has three main unpaired branches
- Other paired branches

Unpaired Branches.....

A. Celiac trunk

- short, thick, branch, which divides into three vessels:

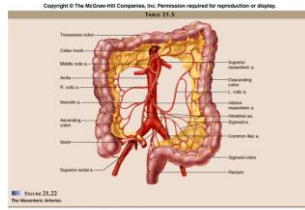
- **Splenic artery**
- supply spleen
- **Left gastric artery**
- supply lesser curvature of the stomach
- **Common hepatic artery**
- supply liver



Cont...

B. Superior mesenteric artery

- supplies small intestine (except portion of duodenum), caecum, appendix, ascending colon and transverse colon

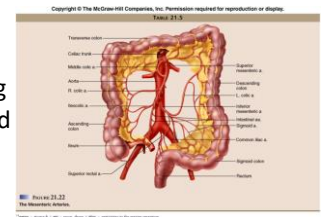


Cont....

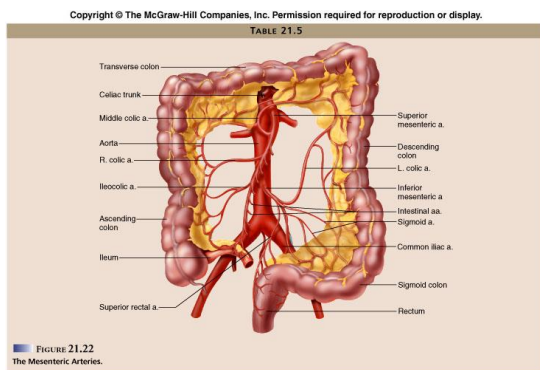
C. Inferior mesenteric artery

- unpaired vessel arise just before bifurcation and supplies descending colon, sigmoid colon and rectum

- Midian sacral artery
- arise at bifurcation and supply sacrum and coccyx.



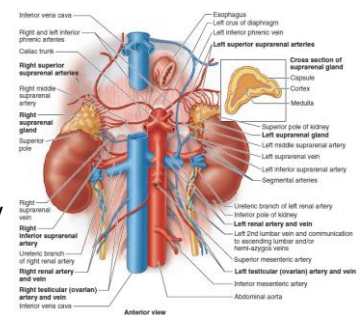
Abdominal blood flow - mesenteric arteries



¹gastric - stomach; ²epi - upon, above; ³plac - pertaining to the greater omentum

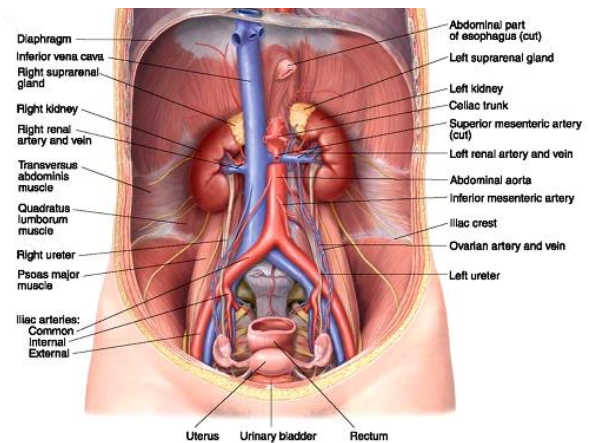
Paired branches.....

- Renal artery
 - to kidney
- Suprarenal artery
 - to adrenal glands
- Testicular artery
 - to testes
- ovarian artery
 - to ovaries



V. Arteries of the pelvis and lower limbs

- The abdominal aorta terminates by bifurcating into
 - right **common** iliac arteries
 - left **common** iliac arteries
- The common iliac divides into
 - the **internal** iliac and
 - **external** iliac.



❖ The internal iliac artery

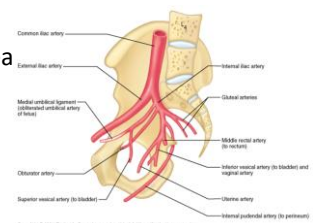
- supply gluteal muscles and organs of the pelvic region
- Urinary bladder, rectum, anal canal.

➤ branches

- **Middle rectal** - Inferior part of rectum, seminal glands, prostate(vagina).
- **Superior, inferior, middle vesicular arteries** - urinary bladder

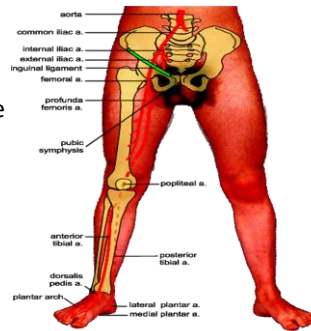
Cont...

- **Uterine and vaginal arteries** - female reproductive organs
- **Superior and inferior gluteal arteries** - glutea muscles.
- **Obturator artery** - upper medial thigh muscles
- **Internal pudendal artery** - perineum and external genitalia of male and female.



The external iliac artery

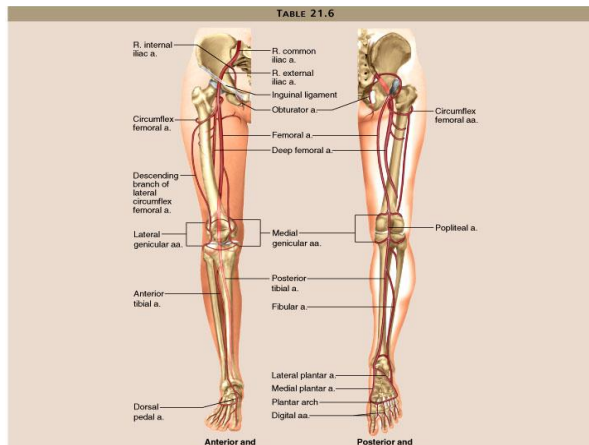
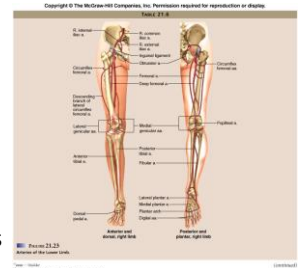
- passes out of pelvis beneath the inguinal ligament to become the **femoral** artery.



➤ Femoral artery

- passes through the femoral triangle on the upper medial portion of the thigh.
- At this point it is close to the surface, hence for **palpation** and **pressure**.
- the femoral artery becomes the **popliteal artery** as it passes across the posterior aspect of the knee.

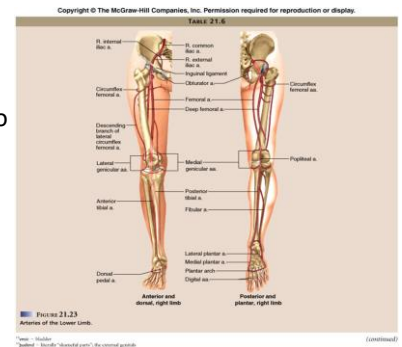
Cont...



Cont...

❖ The popliteal artery

- divides into the
 - anterior tibial
 - and the posterior tibial arteries



➤ the **anterior tibial artery** at the ankle, becomes

- the **dorsal pedal** artery
- forms the plantar arch with the lateral plantar artery of the posterior tibial artery.

➤ The **posterior tibial artery**

- forms the **large fibular artery** which supplies the peroneal muscles of the leg.
- At the ankle it bifurcates into the **lateral** and **medial plantar arteries**.
- The lateral plantar artery forms the plantar arch and gives off digital arteries to the digits of the foot.

Veins of the upper extremity

➤ Consists of superficial and deep venous drainage

▪ Deep veins

- accompany the **arteries** and bear their names / region.
 - radial vein & ulnar vein
 - both drain from deep and superficial palmar arches
 - radial and ulnar veins join in the cubital fossa to form the **brachial** vein, which continues up on the medial side of the arm.
 - Brachial vein -axillary → subclavian → internal jugular → brachiocephalic

Venous drainage of the body

Veins Draining the neck and head

▪ External jugular vein

- from scalp, portions of face, superficial neck region
- drain into right and left subclavian vein.

▪ Internal jugular vein

- from brain, meninges, deep regions of face and neck
- larger and deeper than the external jugular vein

▪ Subclavian vein and internal jugular unite to form the brachiocephalic vein

- the **two brachiocephalic** veins merge to form the superior vena cava, which empties into the right atrium.

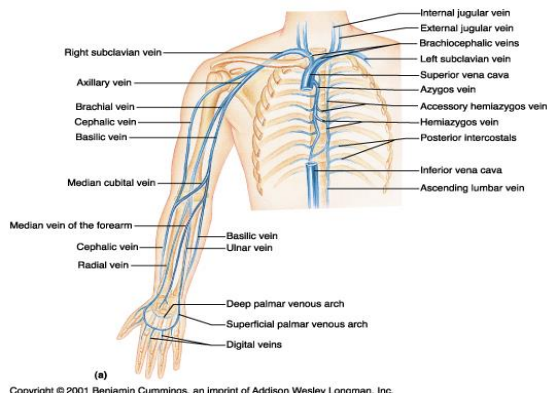
Superficial veins of UL

1. Basilic vein

- drains blood from **ulnar side** of **forearm**, medial side of arm
- merges with **brachial vein** near the head of the humerus to form the **axillary** vein

2. Cephalic vein

- drains superficial region of hand and forearm on **radial side**
- joins axillary vein in the shoulder region
- **median cubital vein** ascends from the cephalic vein to join basilic vein on radial side.
- It is a site of **venipuncture**



Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.

Veins of the thorax

■ Superior vena cava

- receives blood from the right and left **brachiocephalic** veins, which drain head, neck, and upper limb as well as from azygous veins.
- **lacks valves** which are characteristics of most veins

■ The azygous vein

- extends superiorly along the dorsal abdominal and thoracic walls on the right side of the vertebral column.

Cont...

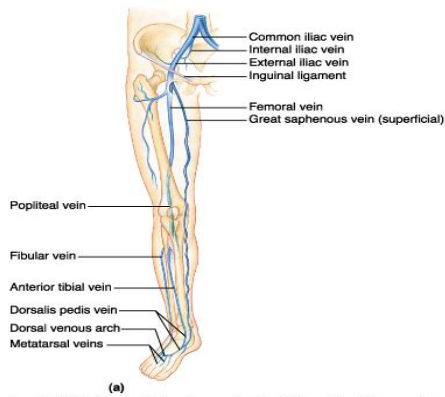
- joins the superior vena cava at T4.
- Its tributaries are:
 - ascending lumbar veins - drain from lumbar and sacral regions
 - Intercostal veins- from intercostals regions
 - accessory hemiazygous and hemiazygous veins- from left of the vertebral column

Veins of the lower extremity

- have a deep and a superficial group
- **The deep veins**
- accompany the corresponding arteries

❖ These include:

- **posterior and anterior tibial veins**
 - originate in the foot and descend upwards in front of the tibia to the back of the knee where they merge to form the popliteal vein.
- **Popliteal vein**
 - drains blood from the knee region and above the knee, it becomes the femoral vein
- **the femoral vein**
 - receives blood from the deep femoral vein and above this, receives from the great saphenous vein, then becomes the external iliac vein (as it passes under the inguinal ligament).
- **the external iliac vein**
 - merges with the internal iliac vein to form the common iliac vein



Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.

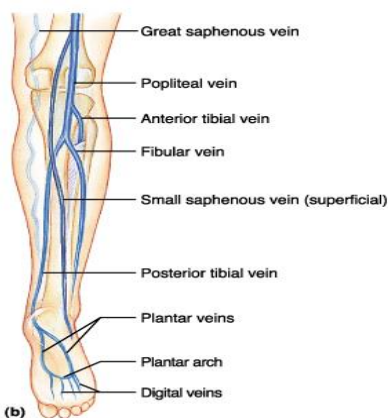
➤ The superficial vein include

▪ small saphenous vein

➤ arises from the **lateral** side of the foot, courses posteriorly along the surface of the calf of the leg and enters deep into the **popliteal** vein behind the knee.

▪ Great saphenous vein

➤ longest vessel in the body, originates at the arch of the foot and ascends superiorly along the **medial** aspect of the leg and thigh before draining into the **femoral** vein.



Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.

Veins of the Abdominal Region

❖ The inferior vena cava

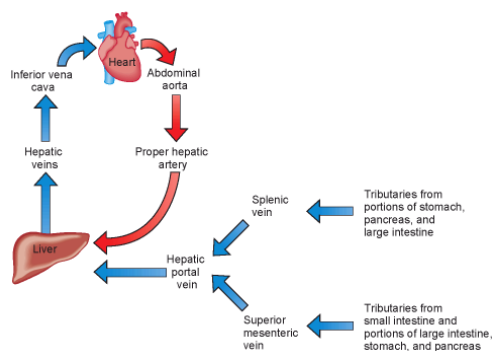
- parallels the abdominal **aorta** on the right as it ascends through the abdominal cavity.
- It penetrates the diaphragm and empties into the **right atrium**
- **largest** in diameter of all vessels in the body
- In the abdomen has tributaries corresponding to the branches of the abdominal aorta .
- (**Exceptions:** the **left testicular** vein, **left ovarian** vein and the **left suprarenal** vein drain into the **left renal vein**)

❖ paired lumbar veins

- renal veins
- right and left testicular veins.
- right and left ovarian veins
- right and left suprarenal veins.
- inferior phrenic vein
- right and left hepatic veins

Hepatic portal system

- **Hepatic portal vein** – drains blood from digestive organs
- formed by union of **superior mesenteric vein** (from small intestine) and **splenic vein** (from spleen)
- **Splenic vein** is formed by convergence of - inferior mesenteric vein (from large intestine), pancreatic vein, left gastroepiploic vein.
- **The right gastroepiploic vein** from stomach drains directly into the superior mesenteric vein.
- Three veins other veins drain directly into the portal vein: **left and right gastric vein** (from lesser curvature) and **cystic vein** (from gall bladder).

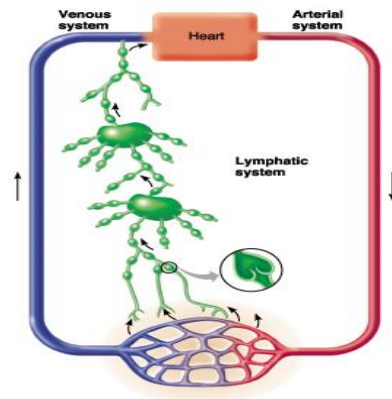


The lymphatic system

✓ is a network of **lymphatic vessels** that returns tissue fluid to the **venous system**.

Function

1. It transports excess **interstitial** (tissue) fluid which was initially formed as a **blood filtrate** back to the blood stream
2. It serves as the route by which an absorbed **fat** from the intestine is **transported** to the blood.
3. It helps provide **immunological** defenses against disease causing agents.



Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.

Lymph and lymph capillaries

- The **smallest** vessels in the lymphatic system are **lymph capillaries**
- The walls are composed of simple squamous epithelium.
- This fluid is formed as a **filtrate** of **plasma** through blood capillaries and is identical in composition to **plasma** except for a **lower protein** concentration
- Once fluid enters the lymphatic capillaries, it is referred to as **lymph**.

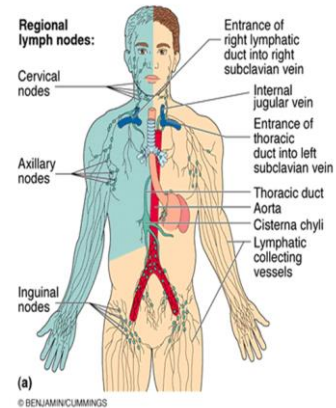
- **Lymph**- a clear, colorless fluid, similar to blood plasma but with much less protein
- May also contain bacteria, viruses, cellular debris or even traveling cancer cells

Lymph vessels

- From capillaries lymph enters into lymph ducts (the wall is similar to vein)
- Lymph ducts eventually empty in to one of the principal vessels
- **Thoracic duct**
- **Right lymphatic duct**

Cont....

- **Thoracic duct** – drain lymph from the lower extremity, abdomen, thoracic region, left upper extremity and left side of head and neck and drains in to left sub clavian vein.
- **Right lymphatic duct** – drains lymph from the right upper extremity , right thoracic region and right side of head and neck and empties in to right sub clavian vein.



Lymph nodes

- masses of B cells and T cells that are surrounded by a capsule.
- Lymph filters through the reticular tissue of lymph nodes
- Lymph nodes are small oval bodies enclosed with in fibrous connective tissue capsules.
- **Afferent** lymph vessels carry lymph to the lymph nodes
- **Efferent** lymph vessels carry lymph away from lymph nodes

Cont...

- Usually occur in clusters in specific regions of the body, some of the principal groups are
- **Popliteal, inguinal nodes**
- **Cubital , axillary nodes**
- **Thoracic and cervical node in the chest and neck respectively**
- **Peyer's patch in the small intestine**

Lymph organs

- Spleen and thymus are lymphoid organs

Spleen

- Spleen is found posterior and lateral to the stomach
- It is not a vital organ in adult. But it assists other body organs in producing lymphocytes, filtering the blood and destroying old erythrocytes.
- In an infant it is an important site for erythrocyte production

Thymus

- Found in the anterior thorax deep to manubrium
- It is much larger in fetus and child than an adult
- Its important site for immunity
- It changes undifferentiated lymphocytes in T lymphocytes
- It houses lymphocytes